The Golden Ratio

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Abstract

The golden ratio $\varphi = (1 + \sqrt{5})/2$ appears in geometry, in number theory and, according to many popular books, almost everywhere else. This note defines the ratio, derives its main algebraic properties, proves that the ratios of consecutive Fibonacci numbers converge to it, and shows how it shapes the golden rectangle. It ends with a warning about the claims that cannot be checked.

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1 Introduction

Few numbers have attracted as much attention as the golden ratio. Euclid described it about 300 BC as the division of a line in "extreme and mean ratio", and later authors gave it names such as the *divine proportion* and the *golden section* [2]. Its popularity today owes much to books that find it in shells, flowers, paintings and buildings [5].

The mathematics behind the ratio is simple and beautiful. In Section 2 we define the ratio and derive the equation it satisfies. Section 3 connects it with the Fibonacci numbers, and Section 4 with a rectangle that can be

cut into a square and a smaller copy of itself. Section 5 separates the facts from the legends.

2 Definition and Basic Properties

Definition 2.1. Two positive numbers $a > b$ are in the *golden ratio* if the ratio of their sum to the larger number equals the ratio of the larger number to the smaller one, that is,

$$\frac{a+b}{a} = \frac{a}{b}. \tag{1}$$

The common value of these ratios is denoted by φ.

Theorem 2.2. *The golden ratio is the positive root of* $x^2 - x - 1 = 0$*, namely*

$$\varphi = \frac{1+\sqrt{5}}{2} \approx 1.6180339887. \tag{2}$$

Proof. Put $x = a/b$, so $x > 1$. Dividing the numerator and the denominator of the left-hand side of (1) by b gives

$$\begin{aligned} \frac{x+1}{x} &= x \\ x+1 &= x^2 \\ x^2 - x - 1 &= 0. \end{aligned}$$

The quadratic formula gives $x = (1 \pm \sqrt{5})/2$. Only the plus sign gives a positive number, which proves (2). □

The other root of the equation is the *conjugate*

$$\psi = \frac{1-\sqrt{5}}{2} \approx -0.6180339887. \tag{3}$$

Since φ and ψ are the roots of $x^2 - x - 1$, the relations between roots and coefficients give

$$\varphi + \psi = 1, \qquad \varphi\psi = -1, \qquad \varphi - \psi = \sqrt{5}. \tag{4}$$

In particular $1/\varphi = -\psi = \varphi - 1 \approx 0.618$: the reciprocal of the golden ratio has the same digits after the decimal point.

Remark. Because $\varphi^2 = \varphi + 1$, every power of φ can be written as $m\varphi + n$ with integers m and n. For example,

$$\varphi^3 = \varphi \cdot \varphi^2 = \varphi^2 + \varphi = 2\varphi + 1, \tag{5}$$

$$\varphi^4 = \varphi \cdot \varphi^3 = 2\varphi^2 + \varphi = 3\varphi + 2. \tag{6}$$

The coefficients $1, 1, 2, 3, \ldots$ already hint at the next section.

The equation $\varphi = 1 + 1/\varphi$ can be applied to itself again and again, which gives the continued fraction and the nested square root

$$\varphi = 1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cdots}}} \qquad \text{and} \qquad \varphi = \sqrt{1 + \sqrt{1 + \sqrt{1 + \cdots}}}.$$

Both expansions use only the digit 1, which makes φ the number that is hardest to approximate by fractions [4].

3 The Fibonacci Numbers

Definition 3.1. The *Fibonacci numbers* are defined by $F_0 = 0$, $F_1 = 1$ and

$$F_{n+1} = F_n + F_{n-1} \qquad (n \geq 1). \tag{7}$$

The sequence begins $0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, \ldots$ Table 1 lists the ratios $\frac{F_{n+1}}{F_n}$ of consecutive terms. They move back and forth around φ and come closer to it at every step.

Table 1: Ratios of consecutive Fibonacci numbers

n	F_n	F_{n+1}	F_{n+1}/F_n	$\lvert F_{n+1}/F_n - \varphi \rvert$
1	1	1	1.000000	0.618034
2	1	2	2.000000	0.381966
3	2	3	1.500000	0.118034
4	3	5	1.666667	0.048633
5	5	8	1.600000	0.018034
6	8	13	1.625000	0.006966
8	21	34	1.619048	0.001014
10	55	89	1.618182	0.000148
12	144	233	1.618056	0.000022

The connection is made exact by a formula published by Binet in 1843, although it was known to de Moivre and Euler more than a century earlier [3].

Lemma 3.2. *Both φ and ψ satisfy $x^{n+1} = x^n + x^{n-1}$ for every $n \geq 1$.*

Proof. Multiply $x^2 = x + 1$ by x^{n-1}. □

Theorem 3.3 (Binet's formula)**.** *For every $n \geq 0$,*

$$F_n = \frac{\varphi^n - \psi^n}{\sqrt{5}}. \tag{8}$$

Proof. Write G_n for the right-hand side of (8). By (4), $G_0 = 0$ and $G_1 = (\varphi - \psi)/\sqrt{5} = 1$. By Lemma 3.2, G_n satisfies the same recurrence (7) as F_n. Two sequences with the same first two terms and the same recurrence are equal, so $G_n = F_n$ for all n. □

Theorem 3.4. *The ratios of consecutive Fibonacci numbers converge to the golden ratio:*

$$\lim_{n\to\infty} \frac{F_{n+1}}{F_n} = \varphi.$$

Proof. Let $r = \psi/\varphi$. Then $|r| = 1/\varphi^2 < 1$, so $r^n \to 0$. By Theorem 3.3,

$$\frac{F_{n+1}}{F_n} = \frac{\varphi^{n+1} - \psi^{n+1}}{\varphi^n - \psi^n} = \varphi \cdot \frac{1 - r^{n+1}}{1 - r^n} \longrightarrow \varphi \cdot \frac{1 - 0}{1 - 0} = \varphi. \qquad \square$$

Since $|\psi| < 1$, the term $\psi^n/\sqrt{5}$ in (8) is smaller than $1/2$ for every $n \geq 1$. Hence F_n is simply the integer nearest to $\varphi^n/\sqrt{5}$; for instance $\varphi^{10}/\sqrt{5} \approx 55.004$ and $F_{10} = 55$ [1].

4 The Golden Rectangle

A rectangle whose sides are in the golden ratio is called a *golden rectangle*. Its most famous property is shown in Figure 1.

Theorem 4.1. *Removing a square from a golden rectangle leaves a golden rectangle.*

Proof. Let the sides be φ and 1. Removing the square of side 1 leaves a rectangle with sides 1 and $\varphi - 1$, and by (4)

$$\frac{1}{\varphi - 1} = \frac{1}{-\psi} = \varphi.$$

So the new rectangle has its sides in the golden ratio. □

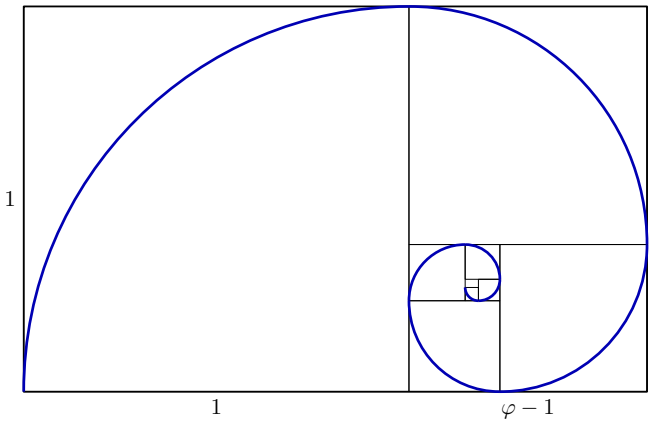


Figure 1: A golden rectangle cut into squares. Each cut leaves a smaller golden rectangle, and the quarter circles form a golden spiral.

Repeating the cut produces the sequence of squares in Figure 1, with sides $1, \varphi^{-1}, \varphi^{-2}, \ldots$ Their areas add up to the area of the rectangle:

$$\sum_{k=0}^{\infty} \varphi^{-2k} = \frac{1}{1-\varphi^{-2}} = \frac{\varphi^2}{\varphi^2-1} = \frac{\varphi^2}{\varphi} = \varphi.$$

Drawing a quarter circle in each square gives the golden spiral, which is often compared with the shell of the nautilus.

5 Facts and Legends

Everything in Sections 2–4 follows from the equation $x^2 = x+1$. Many other claims about the golden ratio do not survive a careful check. Markowsky [6] measured the evidence for the ratio in the Great Pyramid, the Parthenon, paintings and the human body and found most of it to be wishful thinking: with enough freedom in choosing what to measure, almost any ratio near 1.6 can be found. The spiral of the nautilus shell, for example, is logarithmic, but it does not grow by the factor that the golden spiral of Figure 1 would need [5].

The real golden ratio is no less remarkable. It is the limit of the Fibonacci ratios (Theorem 3.4), the number that is hardest to approximate by fractions, and the shape of the only rectangle that reproduces itself when a square is removed (Theorem 4.1).

References

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